Dr. Md Nadim

Curriculum Vitae

Associate Professor
Dept. of CSE, HSTU, Dinajpur-5200, Bangladesh
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Github *Linkedin*

Career Objective

To join a research-intensive postdoctoral position where I can extend my work in quantum and classical machine learning and software engineering, focus on improving software quality and reliability, and contribute to high-impact research, collaboration, and real-world innovation.

Research Interests

- **Empirical Software Engineering & Defect Prediction:** Rigorous evaluation of quality, security, and scalability in human-authored and AI-generated code, with direct implications for industry adoption.
- **Quantum-Enhanced Software Analytics:** Developing hybrid quantum–classical machine learning architectures to solve computationally intensive software engineering problems with improved scalability and performance.

Education

2020–2025: **PhD, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.
Software bug prediction using Quantum and Classical machine learning algorithms, their execution, comparison, and challenges. Utilised IBM Quantum services, D-Wave Quantum Annealing systems, and large language models for experimenting with QML algorithms in real quantum machines and simulators.

2018–2020: **MSc, Computer Science**, *Software Research Lab*, University of Saskatchewan, Canada.

2006–2010 : **BSc, Computer Science & Engineering**, *Hajee Mohammad Danesh Science & Technology University*, Dinajpur, Bangladesh.

Publications

Journal Articles

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Comparative analysis of quantum and classical support vector classifiers for software bug prediction: an exploratory study. *Quantum Machine Intelligence*, volume 7, page 32, Mar 2025.

2025 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Quantum software engineering and potential of quantum computing in software engineering research: a review. *Automated Software Engineering*, volume 32, page 27, Mar 2025.

2023 **Md Nadim** and Banani Roy. Utilizing source code syntax patterns to detect bug inducing commits using machine learning models. *Software Quality Journal*, volume 31, pages 775–807, Sep 2023.

2022 **Md Nadim**, Manishankar Mondal, Chanchal K. Roy, and Kevin A. Schneider. Evaluating the performance of clone detection tools in detecting cloned co-change candidates. *Journal of Systems and Software*, volume 187, page 111229, 2022.

2022 **Md Nadim**, Debajyoti Mondal, and Chanchal K. Roy. Leveraging structural properties of source code graphs for just-in-time bug prediction. *Automated Software Engineering*, volume 29, page 27, Mar 2022.

Conference Proceedings

2025 **Md Nadim**, Mohammad Hassan, Ashis Kumar Mandal, and Chanchal K. Roy. Quantum Vs. Classical Machine Learning Algorithms for Software Defect Prediction: Challenges and Opportunities . In *IEEE/ACM International Workshop on Quantum Software Engineering (Q-SE)*, pages 35–42, Los Alamitos, CA, USA, May 2025. IEEE Computer Society.

2024 Ashis Kumar Mandal, **Md Nadim**, Chanchal K. Roy, Banani Roy, and Kevin A. Schneider. Evaluating the performance of a d-wave quantum annealing system for feature subset selection in software defect prediction. In *IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 02, pages 103–108, 2024.

2020 **Md Nadim**, Manishankar Mondal, and Chanchal K. Roy. Evaluating performance of clone detection tools in detecting cloned cochange candidates. In *IEEE 14th International Workshop on Software Clones (IWSC)*, pages 15–21, 2020.

2016 Shahnaj Parvin Shathi, Md. Delowar Hossain, **Md Nadim**, Sayed Golam Rasul Riayadh, and Tangina Sultana. Enhancing performance of naïve bayes in text classification by introducing an extra weight using less number of training examples. In *2016 International Workshop on Computational Intelligence (IWCI)*, pages 142–147, 2016.

2010 Ashis Kumar Mandal, Md. Delowar Hossain, and **Md Nadim**. Developing an efficient search suggestion generator, ignoring spelling error for high speed data retrieval using double metaphone algorithm. In *13th International Conference on Computer and Information Technology (ICCIT)*, pages 317–320, 2010.

Research Experience

University of Saskatchewan, Canada

Sep, 2018 – Apr, 2025 ***Graduate Researcher***.
Developing a deep multi-modal architecture for accurately predicting protein interaction information from biomedical text.

Advisor : **Dr. Chanchal K. Roy**, *Professor, Department of Computer Science*, University of Saskatchewan, Canada (*Personal Web-page*)

Fellowships & Awards

2018 Dean's Scholarship, University of Saskatchewan

2010 Prime Minister Gold Medal, Bangladesh

Leadership and Community Service

Jan 21, 2024 Crowd Support (Watch Party), TEDx USASK, Canada

2020–2021 Vice President, CSGC, USASK, Canada

2019–2020 GSA Representative, CSGC, USASK, Canada

May 2019 Volunteer Lead (Data Science Workshop), DIGITIZED, USASK, Canada

Key Technical Skills

Programming Python, C, C++, Java

Tools Machine Learning, Deep Learning, Quantum Computing, High-Performance Computing, NumPy, Pandas, Scikit-learn, TensorFlow/PyTorch, Git

Quantum Algorithms Phase Estimation, Amplitude Amplification, Grover's Search, Shor's Algorithm, Hamiltonian Simulation, Qubitization, Quantum Error Correction (QEC)

Web & DB HTML5, JavaScript, PHP, SQL, MySQL

Scripting Linux, Shell Scripting

Certification Courses

2024 **Graduate Professional Skills Certificate @University of Saskatchewan**, *SK*, Canada.
The Graduate Professional Skills Certificate is a comprehensive, non-credit program for graduate students and postdoctoral researchers at the University of Saskatchewan, SK, Canada. Within this Certificate program, GPS 974 focuses on strength-based professional skills development, building reflective practice, and developing a professional portfolio for improving skills and growth.

2024 **Thinking Critically@University of Saskatchewan**, *SK*, Canada.
The goal of this class is to provide a supportive and challenging setting to develop the creative and critical thinking skills required for professional practice. GPS 984 focuses on foundational frameworks of thinking (often invisible to us) that are used for almost everything we do in our personal and professional lives.

Teaching Assistantship @USASK, SK, Canada

Winter, 2024 **CMPT 340: Programming Language Paradigms**, *132 Hours.*
Fall, 2023 **CMPT 214: Programming Principles and Practice**, *88 Hours.*
Spring, 2023 **CMPT 145: Principles of Computer Science**, *66 Hours.*
Winter, 2023 **CMPT 280: Intermediate Data Structures and Algorithms**, *88 Hours.*
Winter, 2022 **CMPT 141: Introduction to Computer Science**, *88 Hours.*
Winter, 2021 **CMPT 470: Advanced Software Engineering**, *88 Hours.*
Fall, 2020 **CMPT 214: Programming Principles and Practice**, *88 Hours.*

Referees

Dr. Chanchal K. Roy
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